Data Engineer

Find the interactive version of this roadmap and more roadmaps at

roadmap.sh

Pre-requisites
- Python Roadmap
- SQL Roadmap

Related Roadmaps
- Data Analyst Roadmap
- AI & Data Scientist Roadmap

Introduction
- What is Data Engineering?
- Data Engineering vs Data Science
- Skills and Responsibilities
- Data Engineering Lifecycle
- Choosing the Right Technologies

Learn the Basics

Python is recommended
- Python
- Java
- Scala
- Go

- Programming Skills
- Data Structures and Algorithms
- Git and GitHub
- Linux Basics
- Networking Fundamentals
- Distributed Systems Basics

Data Engineering Lifecycle

Understand Different Steps
- Data Generation 1
- Data Storage 2
- Data Ingestion 3
- Data Serving 4

Data Generation

Sources of Data
- Database
- APIs
- Logs
- Mobile Apps
- IoT

Data Collection Considerations

Data Storage

- Data Normalization
- Data Modelling Techniques
- CAP Theorem
- OLTP vs OLAP

Database Fundamentals
- Slowly Changing Dimension - SCD
- Horizontal vs Vertical Scaling
- Star vs Snowflake Schema

Relational Databases
- Learn SQL
- Indexing
- Transactions

Relational Databases
- MySQL
- PostgreSQL
- MariaDB
- Aurora DB
- Oracle
- MS SQL

NoSQL Databsases

Document	Column	Graph	Key-Value
MongoDB	Cassandra	Neo4j	Redis
ElasticSearch	BigTable	Neptune	Memcached
CosmosDB	HBase		DynamoDB
CouchDB			

Data Warehousing

What is Data Warehouse?

Data Warehousing Architectures

Data Warehouse
- Google BigQuery
- Snowflake
- Amazon Redshift

Data Lake
- Databricks Delta Lake
- Snowflake
- Onehouse

- Data Mart
- Data Mesh

Other Data Architectures
- Data Fabric
- Data Hub
- Metadata-first Architecture
- Serverless Options

Cloud Computing

Cloud Architectures

Cloud Providers

AWS
- Amazon EC2 (Compute)
- S3 (Storage)
- Amazon RDS (Database)
- Glue (ETL)

Azure
- Azure Virtual Machines
- Azure Blob Storage
- Azure SQL Database
- Data Factory (ETL)

Google Cloud
- Compute Engine (Compute)
- Google Cloud Storage
- Cloud SQL (Database)
- Dataflow

Data Ingestion

Types of Data Ingestion
- Batch
- Hybrid
- Streaming
- Realtime

Data Pipelines

ETL Process
- Extract Data
- Transform Data
- Load Data

Data Pipeline Tools
- Apache Airflow
- dbt
- Luigi
- Perfect

Cluster Computing Basics
- What is Cluster Computing
- Distributed File Systems
  - HDFS
- Job Scheduling
- Cluster Management Tools
  - Kubernetes
  - Apache Hadoop YARN

Big Data Tools
- Apache Spark

Hadoop Ecosystem
- HDFS
- YARN
- MapReduce

Containers & Orchestration
- Docker
- Kubernetes
- Google Cloud GKE
- AWS EKS

CI/CD
- GitHub Actions
- Circle CI
- GitLab CI
- ArgoCD

Monitoring
- Prometheus
- Datadog
- Sentry
- New Relic

Testing
- Unit Testing
- Integration Testing
- End-to-End Testing
- Functional Testing
- A/B Testing
- Load Testing
- Smoke Testing

Messaging Systems
- What and why use them?
- Async vs Sync Communication
- Messages vs Streams
- Best Practices

Common Tools
- Apache Kafka
- RabbitMQ
- AWS SQS
- AWS SNS

Infrastructure as Code - IaC
- Declarative vs Imperative
- Idempotency
- Reusability
- Environmental Management

Common Tools
- Terraform
- OpenTofu
- AWS CDK
- Google Deployment Mgr.

Data Serving

Visit the Data Analyst Roadmap

- Data Analytics
- Business Intelligence

BI Tools
- Microsoft Power BI
- Streamlit
- Tableu
- Looker

Reverse ETL
- ETL vs Reverse ETL
- Reverse ETL Usecases

Tools
- Hightouch
- Census
- Segment

Security

- Authentication vs Authorization
- Encryption
- Tokenization
- Data Masking
- Data Obfuscation

Data Governance

- Data Quality
- Data Lineage
- Metadata Management
- Data Interoperability
- Data Quality

Privacy

Data and AI Regulations
- GDPR
- ECPA
- EU AI Act

- Machine Learning
- MLOps

Also visit the following related roadmaps
- Python
- AI & Data Scientist
- SQL
- Data Analyst
- MLOps